

Do **not** write solutions on this page.

12. [Maximum mark: 21]

- (a) Find $\int (x^2 - 5)e^x dx$. [6]

Consider the differential equation $\frac{dy}{dx} = x^2 - y - 5$.

- (b) By solving the differential equation, show that its solution can be expressed in the form $y = x^2 - 2x - 3 + Ce^{-x}$, where C is a constant. [4]

- (c) Sketch the curve of the particular solution which passes through the point $(-3, 2)$, for $-4 \leq x \leq 4$, clearly labelling the coordinates of any local maximum and minimum points. [5]

Consider the family of curves that are solutions of the differential equation.

The tangent at $x = -3$ is drawn for each of these curves.

- (d) By considering the curve which passes through the point $(-3, p)$ and the curve which passes through the point $(-3, q)$, where $p, q \in \mathbb{R}$, $p \neq q$, show that all these tangents intersect at a common point, and state its coordinates. [6]

